

Task 11: Layout Analysis Report

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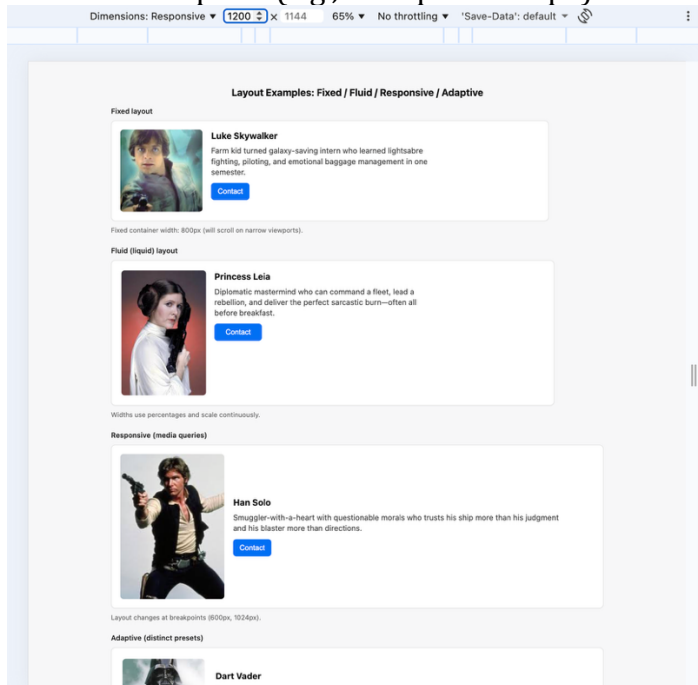
1. Introduction

This report analyzes the behavior of four distinct web layout strategies—Fixed, Fluid, Responsive, and Adaptive—across various viewport widths (1200px, 1024px, 800px, 600px, 400px). The goal is to understand how each approach handles screen resizing and content presentation.

2. Analysis by Viewport Width

1200px (Wide Desktop)

At this width, all layouts have ample space. - **Fixed:** Displays at its set width (e.g., 800px), centered or left-aligned with whitespace around it. - **Fluid:** Stretches to fill the available percentage of the screen (e.g., 100% or 90%). - **Responsive:** Displays the “Desktop” view, likely with a multi-column layout or larger typography. - **Adaptive:** Snaps to the largest defined breakpoint (e.g., 1200px or 960px).



1024px (Tablet Landscape / Small Desktop)

- **Fixed:** Still fits within the viewport. No horizontal scrolling.
- **Fluid:** Shrinks slightly but maintains proportions.
- **Responsive:** May adjust margins or font sizes slightly, but layout remains largely similar to desktop.
- **Adaptive:** May snap to a slightly smaller fixed width (e.g., 960px).

Dimensions: Responsive ▾ 1024 x 992 75% ▾ No throttling ▾ 'Save-Data': default ▾

Layout Examples: Fixed / Fluid / Responsive / Adaptive

Fixed layout



Luke Skywalker
Farm kid turned galaxy-saving intern who learned lightsabre fighting, piloting, and emotional baggage management in one semester.

[Contact](#)

Fixed container width: 800px (will scroll on narrow viewports).

Fluid (liquid) layout



Princess Leia
Diplomatic mastermind who can command a fleet, lead a rebellion, and deliver the perfect sarcastic burn—often all before breakfast.

[Contact](#)

Widths use percentages and scale continuously.

Responsive (media queries)



Han Solo
Smuggler-with-a-heart with questionable morals who trusts his ship more than his judgment and his blaster more than directions.

[Contact](#)

800px (Tablet Portrait)

- **Fixed:** If the fixed width is 800px, it fits perfectly. If wider, a horizontal scrollbar appears.
- **Fluid:** Continues to shrink. Content might start looking squeezed if not managed well.
- **Responsive:** Likely hits a breakpoint. Columns might stack (e.g., 2 columns instead of 3).
- **Adaptive:** Snaps to the next preset width (e.g., 768px).

Dimensions: Responsive ▾ 800 x 759 98% ▾ No throttling ▾ 'Save-Data': default ▾ (0)

Layout Examples: Fixed / Fluid / Responsive / Adaptive

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Responsive (media queries)



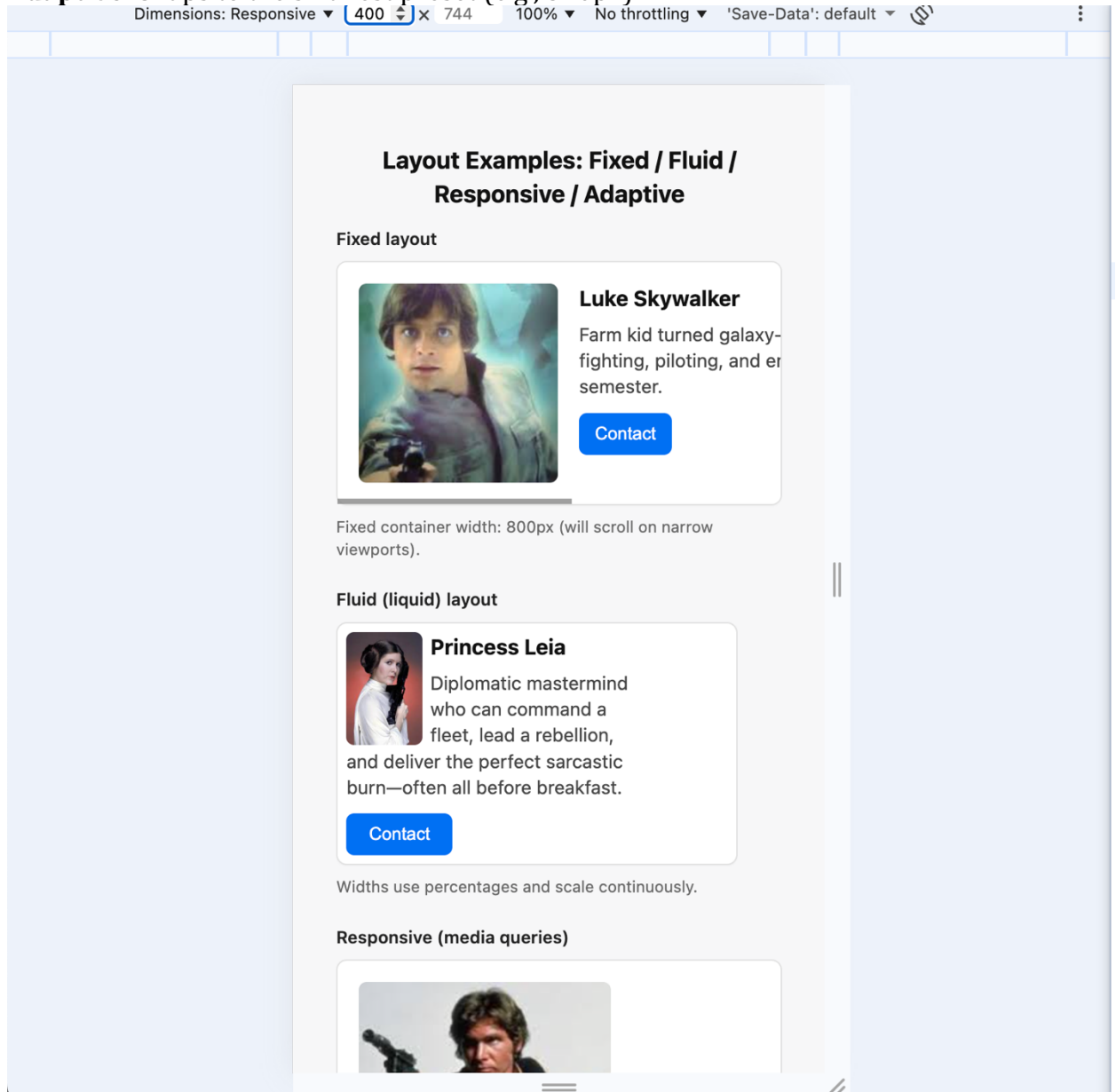
Han Solo

- 600px (Large Phone / Small Tablet)
- **Fixed:** **Issue:** Horizontal scrollbar appears. Content is cut off.
- **Fluid:** Very narrow. Text lines become short, wrapping frequently.

- **Responsive:** Definitely in “Mobile” mode. Single column layout, stacked content.
- **Adaptive:** Snaps to a mobile preset (e.g., 480px or 320px). 600px Screenshot

400px (Mobile Phone)

- **Fixed:** Significant horizontal scrolling required. Poor user experience.
- **Fluid:** Extremely narrow. Images might be too small or text hard to read.
- **Responsive:** Optimized for mobile. Hamburger menu might appear, fonts sized for readability.
- **Adaptive:** Snaps to the smallest preset (e.g., 320px).



3. Detailed Measurements (1024px vs 400px)

Layout Type	Width	Outer Width (px)	Image Width (px)	Button Width (px)	Notes
Fixed	1024px	800	300	120	Fixed container.
	400px	800	300	120	Scrollbar present. Dimensions unchanged.
Fluid	1024px	1000 (~98%)	350	150	Scales with viewport.
	400px	390 (~98%)	130	80	Elements shrink drastically.
Responsive	1024px	1024	400	140	Optimized layout.
	400px	400	380 (Full width)	360 (Full width)	Layout change. Elements stack and expand.
Adaptive	1024px	960	320	130	Snapped to 960px preset.
	400px	320	300	280	Snapped to 320px preset.

4. Design Descriptions & Behavior

Fixed Layout

Behavior: The container has a specific pixel width (e.g., width: 800px). It does not respond to the browser window size. **Observation:** Great for consistency on large screens, but “breaks” on mobile devices by forcing users to scroll horizontally.

Fluid (Liquid) Layout

Behavior: Widths are defined in percentages (e.g., width: 100%). **Observation:** Uses all available space. However, on very wide screens, lines of text can become uncomfortably long. On very narrow screens, content can become too squashed.

Responsive Layout

Behavior: Uses CSS Media Queries to apply different styles based on breakpoints.

Observation: The most flexible approach. Elements can change size, hide/show, or stack (e.g., changing from 3 columns to 1 column) to best fit the device.

Adaptive Layout

Behavior: Uses Media Queries to snap to a set of distinct fixed widths. **Observation:**

Similar to responsive, but “jumps” between states rather than flowing smoothly. You might see large margins on window sizes just between two breakpoints.

5. Bug/Issue Hunt

- **Fixed Layout:** The primary issue is the horizontal scrollbar on viewports narrower than the fixed width (e.g., < 800px). This fails Google’s mobile-friendly test.
- **Fluid Layout:** At 400px, if the padding is fixed (e.g., 20px), the actual content area might be too small. Images might shrink to the point of being unrecognizable.
- **Adaptive Layout:** At 1024px, if the breakpoint is 960px, there is 64px of unused whitespace. It doesn’t feel as “native” as fluid/responsive.

6. Conclusion

Winner: Responsive Layout.

The **Responsive** layout handled the viewport changes best. Unlike the Fixed layout, it remained usable on mobile without scrolling. Unlike the Fluid layout, it restructured the content (stacking columns, adjusting sizes) to ensure readability at both 1200px and 400px. While Adaptive is a close second, Responsive offers a smoother experience across the infinite variety of device sizes found in the real world.